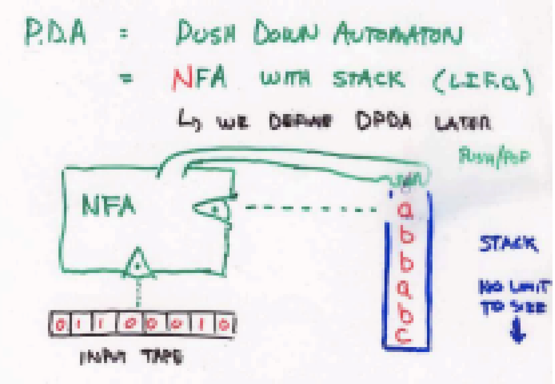
PDA:a NFA coupled

BFA/NFA/RE

Constant

memory



Computation

NFA + stack -

Context-free grammar

PDAs and CFGs are equivalent: both generate exactly CFLs.
PDA is a machine-centric view of CFLs.

PDA

Non deterministic Events Cannot be determined precisely

Pushdown : Using Stack For infinite memory

there

Automaton Computing machine For one situation, Can be multiple possible

↑ Finite Set /set of States) S: & x([∀(33) x [redacted] x **) = input alphabet

A PDA P is 6 tuple

M = (1, E, T, s, 90, F)

transition 3

2: Start state

To Finite set [stack alphabet) F: Final Set of Final States

= $) Di x)

A PDA M = (2, 5, 3, %, 5) accepts a string ~~WEI~~* ifo [redacted]

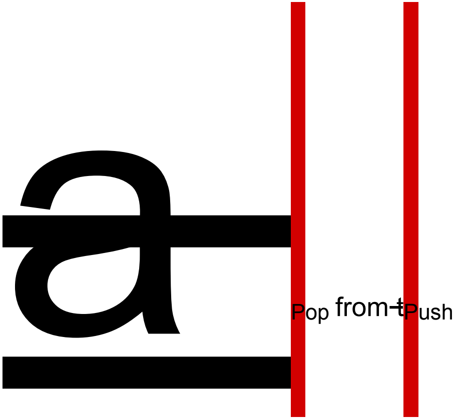
190, w

Initial Configuration

For some ~~hert~~ and ~~somenal~~ configuration

$\vdash^*$ means Yields in a finite

$\vdash$ yields in one more Sequence of moves of Machine M.



Pop
from
Push

① = ~~&Wesaiby~~ * ~~and~~ ~~nee~~ n =3

L = &E . ab . aabb---]

Transition : read an alphabet, Stack From Stack

Push a $ in the ~~beginning~~. ~~If~~ we can pop a $ the stack is empty.

~~DonD~~

#P#A

9 2 Y b x E

~~&~~ ~~wed~~

w is a

~~a b~~ ~~a~~

A ~~palindrome~~ aba ~~aaa~~

When reading

~~When~~ readingba ~~pop~~ o

read, pop push

~~is~~ of if

Stack

~~b~~,2 -3A

~~O~~

a=~~69~~

9, E~~E~~E ~~15~~ b 2 - -E

③ 1 = ~~\Wes0~~ z * yun ; ~~N~~ ~~&~~

n $^=$2 ~~*81111~~ Read + push two X Read, Pop [redacted] us [redacted]

~~1: X~~ [redacted] E

~~Dosef~~

8, 2 x)83 '204

[redacted] * Gwesozy * y3n2 nN = [2

80011 0000001111]

[redacted]

x = E

[redacted]

n=y M = 1 n = 2

[redacted]ad t[redacted]ree J's - push t[redacted]o x's

E D [redacted] 303 = $388 E

by33

&x8

3,5 X

31,

$\geq$ 280

⑤ 1 :SWES0 ~~1~~3 *: ~~gam~~ n > m and ~~neN]~~ ~~8~~, ~~EEX~~

~~4~~ X -E ~~IS~~

m = 2 ~~Odornt~~ ~~of~~ ~~s~~ and

n = 3

~~&~~

~~50111~~

8

[abiej or

b's ~~ike~~

[redacted]

[redacted] E

c = 2 2

[redacted]

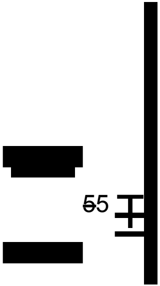
any ~~any~~

amount ~~a~~

~~jaabb~~ ~~c~~,

i =

C^{2+} E | * ; 2 c, x S


55

-E & 1

i K [redacted]

[redacted]

5,2

~~oat~~ d

~~25~~

* L = ~~&WEE~~ */the number of o's in w is not same as the number of 1s Y

1 Pop X ~~P101001~~ : Push

Two possibility Case Amount of 0 ~~71 5~~

Extra x in Stack mean ~~831~~

case 2 Amount of ~~1~~) 3 1 s ~~1 1 18~~ Extra y in stack mean 17 a

*

$>$

$=$

~~amoun~~[redacted] of

~~equal~~ a and ls 8) 1 1 81110 ~~1)~~

~~A~~

~~3~~, 3 -2 * ~~d~~ ~~D~~y

Casan

t

2 E

Eit[redacted]e non-deterministically [redacted]ess which

t[redacted]ansition to take t[redacted]ma[redacted]h o's and

Case1 Cas2 Is here

* = 90 1] 1 5X 350 13 *: X#y the number of oo

substring in X is equal As number of 11 substring in y]

X# Y

① 101110111 ↓

3rd ~~substring~~so

~~D~~

1 ~~St~~ ~~substring~~ ~~of~~ of

2nd ~~substring~~ s~~to~~p

= l#s m ~~amount~~ ~~of~~

amoun~~t~~ o~~f~~

~~11~~ substring ~~of~~ substring

*

[redacted]

[redacted]

~~11~~ ~~X~~ ~~Pop~~ ** z # & > x ~~push~~

~~1000110111~~

[redacted]

E

[redacted]

1 E 8 27X &, S 1 x

A

1 E

~~10~~ # -E

8R , Containe equal number of occurence of substring ab and ba.

~~&~~ 1 = GWE~~d~~a by* /W starts and ends with Same

characters ~~We~~ don't ~~need~~ ~~to~~ ~~Push(POP~~ ~~For~~

Regular ~~lang.~~

h

a T 9 -:[E

Deterministic PDA | #on-deterministic PDA

~~b~~-, ~~22~~

;2-E , se ~~apa~~

~~&~~ b, 2-2

[redacted]

[redacted]

[redacted] ~~[b], see 9~~

[redacted]

ƀ,^{2} Ē Đ

$\geq$

[,! ! [(,! ! (

!,! ! $ *!,* $ *! !*

q_0 q_1 q_2

], [*! !*),(*! !*